

2024 AMC 10B Problem 4 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 4

**Problem:**

Balls numbered $1, 2, 3, \dots$ are deposited in 5 bins, labeled A, B, C, D, and E, using the following procedure. Ball 1 is deposited in bin A, and balls 2 and 3 are deposited in bin B. The next 3 balls are deposited in bin C, the next 4 in bin D, and so on, cycling back to bin A after balls are deposited in bin E. (For example, balls numbered 22, 23, ... 28 are deposited in bin B at step 7 of this process.) In which bin is ball 2024 deposited?

Answer Choices:

- A. A
- B. B
- C. C
- D. D
- E. E

Solution:

After n steps, a total of $1 + 2 + 3 + \dots + n = \frac{1}{2}n(n + 1)$ balls have been deposited. In particular, after step $n = 63$, a total of $\frac{1}{2} \cdot 63 \cdot 64 = 2016$ balls have been deposited. The next batch of 64 balls will include ball 2024. Because 64 has remainder 4 when divided by 5, ball 2024 is deposited in $\text{bin}(D)$ D .

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